

CNN and RNN

CSE427: Machine Learning

1 Part A: CNN Theory Questions

Practice Questions

1. What is the main motivation for using CNNs instead of fully connected neural networks for image data?
2. Explain why a fully connected network has too many parameters for images.
3. What is meant by local connectivity in CNNs?
4. What is meant by parameter sharing in CNNs?
5. Explain the difference between an input image, a filter, and a feature map.
6. What is the role of a filter in a CNN?
7. Why can a filter be interpreted as a feature detector?
8. Give examples of features that early CNN filters may detect.
9. What is the difference between a grayscale image and an RGB image in terms of tensor shape?
10. If an RGB image has shape $227 \times 227 \times 3$, what does the depth 3 represent?
11. Why must the depth of a convolution filter match the depth of the input image?
12. If the input image has depth 3, why does a filter have depth 3?
13. What is a feature map?
14. Why does one filter produce one feature map?
15. If a convolution layer has K filters, how many feature maps will it produce?
16. Why does the output depth of a convolution layer equal the number of filters?
17. What is zero padding?
18. Why do we use padding in CNNs?
19. What is stride?
20. How does increasing stride affect the output size?
21. What is same padding?
22. What is valid padding?
23. Why are odd-sized filters such as 3×3 , 5×5 , and 7×7 commonly used?
24. What is pooling?
25. What is max pooling?
26. What is average pooling?
27. Compare max pooling and average pooling.
28. Why does pooling reduce computation?
29. Why does pooling provide some translation robustness?
30. What is flattening in CNNs?
31. Why do we need flattening before a dense layer?
32. Why can fully connected layers dominate the parameter count in CNN architectures?
33. What is normalization?
34. Why is normalization useful during CNN training?
35. Compare Batch Normalization, Layer Normalization, Instance Normalization, and Group Normalization.
36. Why may Batch Normalization perform poorly with very small batch sizes?
37. Why is Layer Normalization independent of batch size?
38. What is the main design idea of VGGNet?
39. Why does VGGNet use repeated 3×3 convolution layers?
40. In VGGNet, why does 3×3 convolution with stride 1 and padding 1 keep spatial size unchanged?

2 Part B: CNN Formula Sheet

Important Formula

For input:

$$W_1 \times H_1 \times D_1$$

filter size:

$$F \times F \times D_1$$

number of filters:

$$K$$

stride:

$$S$$

padding:

$$P$$

the output size is:

$$W_2 = \frac{W_1 - F + 2P}{S} + 1$$

$$H_2 = \frac{H_1 - F + 2P}{S} + 1$$

$$D_2 = K$$

So:

$$\boxed{Output = W_2 \times H_2 \times K}$$

Important Formula

Number of trainable parameters in a convolution layer:

$$\boxed{(F \times F \times D_1) \times K + K}$$

where:

$$(F \times F \times D_1) \times K$$

is the number of weights, and:

$$K$$

is the number of biases.

3 Part C: CNN Mathematical Problems

Practice Questions

1. An input image has size $64 \times 64 \times 3$. A convolution layer uses 16 filters of size 5×5 , stride 1, and padding 2. Find:
 - (a) output width;
 - (b) output height;
 - (c) output depth;
 - (d) total output dimension;
 - (e) number of trainable parameters.
2. An input image has size $32 \times 32 \times 3$. A convolution layer uses 10 filters of size 3×3 , stride 1, and padding 1. Find the output dimension and number of parameters.
3. An input image has size $227 \times 227 \times 3$. A convolution layer uses 90 filters of size 11×11 , stride 4, and padding 0. Find:
 - (a) output width;
 - (b) output height;
 - (c) output depth;
 - (d) total number of output activations;
 - (e) trainable parameters.
4. An input feature map has size $28 \times 28 \times 16$. A convolution layer uses 32 filters of size 3×3 , stride 1, and padding 1. Find the output size and parameter count.
5. An input feature map has size $28 \times 28 \times 16$. A convolution layer uses 32 filters of size 3×3 , stride 1, and padding 0. Find the output size and parameter count.
6. Compare the answers of Question 4 and Question 5. What effect did padding have?
7. An input has size $128 \times 128 \times 3$. A convolution layer uses 64 filters of size 7×7 , stride 2, and padding 3. Find the output dimension.
8. An input has size $100 \times 100 \times 3$. A convolution layer uses 20 filters of size 5×5 , stride 2, and padding 0. Find the output dimension.
9. An input has size $50 \times 50 \times 8$. A convolution layer uses 12 filters of size 3×3 , stride 1, and padding 1. Find the output dimension and trainable parameters.
10. An input has size $16 \times 16 \times 32$. A 1×1 convolution layer uses 64 filters. Find:
 - (a) output dimension;
 - (b) parameter count.
11. Explain why 1×1 convolution can change the depth of a feature map without changing spatial size.
12. A 4×4 feature map is given:

$$X = \begin{bmatrix} 1 & 3 & 2 & 4 \\ 5 & 6 & 1 & 2 \\ 0 & 1 & 7 & 8 \\ 3 & 2 & 5 & 9 \end{bmatrix}$$

- Apply 2×2 max pooling with stride 2. Show the output.
13. For the same feature map in Question 12, apply 2×2 average pooling with stride 2. Show the output.
 14. An input feature map has size $64 \times 64 \times 16$. After 2×2 max pooling with stride 2, what is the output dimension?
 15. An input feature map has size $32 \times 32 \times 64$. After 2×2 max pooling with stride 2, what is the output dimension?
 16. An input image has size $224 \times 224 \times 3$. A VGG-style convolution layer uses 64 filters of size 3×3 , stride 1, and padding 1. Find the output dimension and number of parameters.
 17. The output of a convolution block is $7 \times 7 \times 512$. After flattening, how many values are in the vector?

4 Part D: CNN Matrix Multiplication Questions

Practice Questions

1. For an input image of size $227 \times 227 \times 3$, filter size $11 \times 11 \times 3$, stride 4, and padding 0, the output size is 55×55 . How many local image blocks are there?
2. Each local block has size $11 \times 11 \times 3$. How many values are in each block?
3. If each block is converted into one column, what is the size of the image block matrix?
4. If there are 90 filters, each with 363 weights, what is the size of the filter matrix?
5. Multiply the filter matrix and image block matrix:

$$(90 \times 363)(363 \times 3025)$$

What is the output matrix size?

6. Explain how the matrix:

$$90 \times 3025$$

can be reshaped into:

$$55 \times 55 \times 90$$

7. Why is convolution sometimes implemented as matrix multiplication?
8. What is the relationship between `im2col` and convolution?

5 Part E: CNN Conceptual Short Answers

Practice Questions

1. Explain in words why CNNs have fewer parameters than fully connected networks.
2. Explain why one filter can detect the same feature at different spatial locations.
3. Why is parameter sharing useful for image recognition?
4. Why does a CNN preserve spatial structure better than a fully connected network?
5. Why does padding help border pixels contribute more?
6. What happens if we use no padding in many convolution layers?
7. What happens if stride is increased?
8. What is the difference between output feature map and filter?
9. Why do multiple filters produce multiple channels in the output?
10. Why does max pooling keep the strongest activation?
11. In backpropagation through max pooling, which input receives the gradient?
12. Why does average pooling distribute information more smoothly?
13. Why are normalization layers useful in deep CNNs?
14. Why do modern CNNs often avoid very large fully connected layers?
15. Explain the phrase: "CNNs have many output activations but relatively few shared weights."

6 Part F: RNN Theory Questions

Practice Questions

1. What is sequential data?
2. Give five examples of sequential data.
3. Why are ordinary feedforward neural networks not ideal for sequential data?
4. What is the main idea of a Recurrent Neural Network?
5. What is the feedback loop in an RNN?
6. Why does an RNN have memory?
7. What is a hidden state?
8. What is the difference between input x_t , hidden state h_t , and output y_t ?
9. Write the basic RNN recurrence equation.
10. Explain the meaning of:

$$h_t = f(W_{xh}x_t + W_{hh}h_{t-1} + b_h)$$

11. What is meant by unrolling an RNN?
12. Why do we unroll an RNN?
13. Does unrolling create new trainable weights? Explain.
14. What does it mean that RNN weights are shared across time?
15. Why can an RNN handle different sequence lengths?
16. If Company A has 9 days of stock data and Company B has 5 days of stock data, how does the same RNN process both?
17. What is many-to-one RNN?
18. Give examples of many-to-one RNN tasks.
19. What is many-to-many RNN?
20. Give examples of many-to-many RNN tasks.
21. What is autoregressive generation?
22. How is autoregressive generation different from many-to-many prediction?
23. What is part-of-speech tagging?
24. Why is part-of-speech tagging a many-to-many task?
25. What is Backpropagation Through Time?
26. Why is BPTT needed in RNNs?
27. Why does the same weight affect many time steps?
28. What is the vanishing gradient problem?
29. What is the exploding gradient problem?
30. Why do vanishing and exploding gradients occur in RNNs?
31. Why do basic RNNs struggle with long-term dependencies?
32. What are LSTM and GRU?
33. Why were LSTM and GRU introduced?
34. What is gradient clipping?
35. How does gradient clipping help with exploding gradients?

7 Part G: RNN Formula Sheet

Important Formula

Basic RNN recurrence:

$$h_t = f(W_{xh}x_t + W_{hh}h_{t-1} + b_h)$$

Output:

$$y_t = g(W_{hy}h_t + b_y)$$

Important Formula

Simple scalar RNN used for practice:

$$h_t = x_t W_1 + h_{t-1} W_2$$

where:

W_1 = input weight

W_2 = recurrent weight

Important Formula

One-input case:

$$Predicted = Today \times W_1$$

Two-input case:

$$Predicted = (Yesterday \times W_1 \times W_2) + (Today \times W_1)$$

Three-input case:

$$Predicted = (D_{-2}W_1W_2^2) + (D_{-1}W_1W_2) + (D_0W_1)$$

8 Part H: RNN Mathematical Problems

Practice Questions

1. Suppose

$$h_t = x_t W_1 + h_{t-1} W_2, \quad W_1 = 1, \quad W_2 = 0.5, \quad h_0 = 0.$$

If $x_1 = 0.5$, find h_1 .

2. For the same RNN, if

$$x_1 = 0.5, \quad x_2 = 1,$$

find h_1 and h_2 .

3. For the same RNN, if

$$x_1 = 1, \quad x_2 = 0.5, \quad x_3 = 1,$$

find h_1, h_2, h_3 .

4. Let

$$W_1 = 1.8, \quad W_2 = -0.5, \quad h_0 = 0.$$

For $x_1 = 0, x_2 = 0$, compute h_1, h_2 .

5. Let

$$W_1 = 1.8, \quad W_2 = -0.5, \quad h_0 = 0.$$

For $x_1 = 0, x_2 = 0.5$, compute h_1, h_2 .

6. Let

$$W_1 = 1.8, \quad W_2 = -0.5, \quad h_0 = 0.$$

For $x_1 = 1, x_2 = 0.5$, compute h_1, h_2 .

7. Let

$$W_1 = 1.8, \quad W_2 = -0.5, \quad h_0 = 0.$$

For $x_1 = 1, x_2 = 1$, compute h_1, h_2 .

8. Suppose

$$h_t = \text{ReLU}(x_t W_1 + h_{t-1} W_2), \quad W_1 = 1.8, \quad W_2 = -0.5, \quad h_0 = 0.$$

Repeat Questions 4–7 using ReLU.

9. For the two-input case, derive

$$\text{Predicted} = (\text{Yesterday} \times W_1 \times W_2) + (\text{Today} \times W_1).$$

10. For the three-input case, derive

$$\text{Predicted} = (D_{-2} W_1 W_2^2) + (D_{-1} W_1 W_2) + (D_0 W_1).$$

11. For four inputs

$$D_{-3}, D_{-2}, D_{-1}, D_0,$$

derive the predicted value in terms of W_1 and W_2 .

12. For five inputs

$$D_{-4}, D_{-3}, D_{-2}, D_{-1}, D_0,$$

derive the predicted value.

13. In general, what power of W_2 multiplies D_{-k} ?

14. Suppose $W_2 = 0.5$. Compute

$$W_2^1, \quad W_2^2, \quad W_2^3, \quad W_2^{10}, \quad W_2^{20}.$$

15. Suppose $W_2 = 2$. Compute

$$W_2^1, \quad W_2^2, \quad W_2^3, \quad W_2^{10}, \quad W_2^{20}.$$

Practice Questions

17. Explain which of Questions 14 and 15 leads to vanishing gradient and which leads to exploding gradient.

18. Suppose $W_2 = 0.8$. Compute

$$W_2^5, \quad W_2^{10}, \quad W_2^{20}.$$

Does the gradient shrink or grow?

19. Suppose $W_2 = 1.2$. Compute

$$W_2^5, \quad W_2^{10}, \quad W_2^{20}.$$

Does the gradient shrink or grow?

20. For one-input prediction,

$$\text{Predicted} = \text{Today} \times W_1, \quad \text{SSR} = (\text{Observed} - \text{Predicted})^2,$$

derive

$$\frac{d\text{SSR}}{dW_1}.$$

21. For

$$\text{Observed} = 1, \quad \text{Today} = 0.5, \quad W_1 = 1,$$

compute Predicted , SSR , and $\frac{d\text{SSR}}{dW_1}$.

22. If $\eta = 0.1$, update W_1 using gradient descent for Question 20.

23. For two-input prediction,

$$\text{Predicted} = (\text{Yesterday} \times W_1 \times W_2) + (\text{Today} \times W_1),$$

derive

$$\frac{d\text{Predicted}}{dW_1}.$$

24. For the same two-input prediction, derive

$$\frac{d\text{Predicted}}{dW_2}.$$

25. For

$$\text{Yesterday} = 1, \quad \text{Today} = 0.5, \quad W_1 = 1, \quad W_2 = 0.5,$$

compute Predicted .

26. For Question 24, if $\text{Observed} = 1.5$, compute SSR .

27. For three-input prediction,

$$\text{Predicted} = D_{-2}W_1W_2^2 + D_{-1}W_1W_2 + D_0W_1,$$

derive

$$\frac{d\text{Predicted}}{dW_1}.$$

28. For the same three-input prediction, derive

$$\frac{d\text{Predicted}}{dW_2}.$$

29. Suppose

$$D_{-2} = 1, \quad D_{-1} = 0.5, \quad D_0 = 1,$$

$$W_1 = 1, \quad W_2 = 0.5.$$

Compute the predicted value.

30. Suppose an RNN is unrolled 20 steps and $W_2 = 0.5$. What factor multiplies the oldest input?

31. Suppose an RNN is unrolled 20 steps and $W_2 = 2$. What factor multiplies the oldest input?

32. Explain why Question 29 shows vanishing gradient.

33. Explain why Question 30 shows exploding gradient.

9 Part I: BPTT Theory and Derivation Questions

Practice Questions

1. Explain why the one-input RNN case behaves like a simple single-neuron network.
2. In the one-input case, why is the recurrent connection ignored?
3. Explain the path:

$$Today \rightarrow W_1 \rightarrow Predicted$$

4. Explain the path:

$$Yesterday \rightarrow W_1 \rightarrow W_2 \rightarrow Predicted$$

5. In the two-input case, why does yesterday pass through W_2 , but today does not?
6. In the three-input case, why does day-before-yesterday pass through W_2^2 ?
7. Explain why older inputs are multiplied by higher powers of W_2 .
8. What does it mean to backpropagate through time?
9. Why is BPTT just ordinary backpropagation applied to an unrolled RNN?
10. Why does one shared weight receive gradient contributions from multiple time steps?
11. Explain why:

$$\frac{dSSR}{dW_1} = \frac{dSSR}{dPredicted} \cdot \frac{dPredicted}{dW_1}$$

12. If:

$$SSR = (Observed - Predicted)^2$$

show that:

$$\frac{dSSR}{dPredicted} = -2(Observed - Predicted)$$

13. If:

$$Predicted = Today \times W_1$$

show that:

$$\frac{dPredicted}{dW_1} = Today$$

14. Explain why the gradient becomes small when W_2 is repeatedly multiplied and $|W_2| < 1$.
15. Explain why the gradient becomes large when W_2 is repeatedly multiplied and $|W_2| > 1$.
16. What happens when $|W_2| \approx 1$?
17. Why does vanishing gradient make it difficult to learn long-term dependency?
18. Why does exploding gradient make training unstable?
19. How can gradient clipping help?
20. How can LSTM or GRU help compared to basic RNN?

10 Part J: RNN Application Questions

Practice Questions

1. Classify each task as many-to-one, many-to-many, or autoregressive:

- (a) sentence sentiment classification;
- (b) part-of-speech tagging;
- (c) stock prediction from last 10 days;
- (d) next-word generation;
- (e) speech frame labeling;
- (f) machine translation.

2. Explain why sentiment classification is many-to-one.

3. Explain why part-of-speech tagging is many-to-many.

4. Explain why text generation is autoregressive.

5. For the sentence:

Janet will back the bill

assign possible part-of-speech tags.

6. In POS tagging, why does each word need an output label?

7. In autoregressive generation, why is the predicted token used again as input?

8. Give an example of an autoregressive sequence:

$$x_1 \rightarrow x_2 \rightarrow x_3 \rightarrow x_4$$

9. Suppose an RNN reads:

I love machine

and predicts:

learning

Explain the autoregressive process.

10. Explain why RNNs are useful for time-series forecasting.

11 Part K: Mixed CNN and RNN Questions

Practice Questions

1. Compare CNN and RNN in terms of input type.
2. Compare CNN and RNN in terms of parameter sharing.
3. CNNs share weights across spatial positions. RNNs share weights across time steps. Explain.
4. In CNNs, what is a feature map? In RNNs, what is a hidden state?
5. In CNNs, what does a filter learn? In RNNs, what does the recurrent weight learn?
6. Why are CNNs suitable for images?
7. Why are RNNs suitable for sequences?
8. Explain the analogy:

CNN filter slides over space

RNN cell unrolls over time
9. A CNN has many output activations but few shared weights. Explain.
10. An RNN processes many time steps but uses the same weights. Explain.
11. Which model is more appropriate for image classification: CNN or RNN? Explain.
12. Which model is more appropriate for sentence processing: CNN or RNN? Explain.
13. What is the role of backpropagation in CNN?
14. What is the role of BPTT in RNN?
15. Why do RNNs face vanishing/exploding gradient problems more directly than shallow CNNs?
16. What is the difference between spatial dependency and temporal dependency?
17. Give one real-world task where CNN and RNN can be combined.
18. Example: video classification uses CNN and RNN. Explain how.
19. Example: image captioning uses CNN and RNN. Explain how.
20. Write a short paragraph comparing CNN, RNN, and MLP.